Nonnegative and positive factorizations

Polynomial-time minimum-volume decision under sufficient scattering

Difficulty: challenging **Importance:** interesting to the community

Status: Open

Rating rationale: Challenging because the SSC promise gives identifiability without a known polynomial optimization method; community importance reflects minimum-volume factorization in source separation and data analysis.
Topic: structured matrix factorization; global optimization

Reviewed partial progress — 2026-09-13

Sidney Holden (Center for Computational Biology, Flatiron Institute, Simons Foundation) supplies round-two partial results, with verified affiliation and source provenance.

Theorem 2.3 gives conditional exact factor recovery using an exact decision/value oracle on the stronger SSC subclass. Theorems 3.1–3.2 show weak-SSC diagonal rigidity and a punctured interval of projective probes outside the entire original promise. Theorem 4.5 and Corollary 4.7 give all-degree input-preordering SOS obstructions and exponential sizes for standard dense levels. These results neither supply the missing oracle nor exclude all polynomial algorithms, general SDPs or sparse lifts.

A separate independent informal Codex AI-agent review checks the stated partial scopes. The full rational-input decision problem below remains open; no full resolution, external human peer review or formal verification is asserted. No Lean verification was performed. The original weaker SSC promise and computational model are unchanged.

Context and notation

For $H \in \mathbb{R}_+^{r\times n}$, let $\operatorname{cone}(H) = \{Hy : y \in \mathbb{R}_+^n\}$ and let e_d be the all-ones vector in $\mathbb{R}^d$. Here **SSC** means the precise version in Gillis's Definition 4.15:

$$\mathcal{C}_r = \{x \in \mathbb{R}_+^r : e_r^T x \geq \sqrt{r-1}\|x\|_2\} \subseteq \operatorname{cone}(H),$$

and every orthogonal $Q \in \mathbb{R}^{r\times r}$ satisfying $\operatorname{cone}(H) \subseteq \operatorname{cone}(Q)$ is a permutation matrix. Some later sources use a stronger second condition involving the boundary of the dual cone; that condition is not substituted here.

For $X \in \mathbb{R}^{m\times n}$ and $r = \operatorname{rank}(X)$, the minimum-volume problem considered here is

$$\min_{W\in\mathbb{R}^{m\times r},\, H\in\mathbb{R}_+^{r\times n}} \det(W^TW) \quad \text{subject to} \quad X = WH, \qquad e_r^T H = e_n^T. \tag{MV}$$

In particular, W is unrestricted in sign and the **columns** of H sum to one. This is min-vol NMF (1), Definition 4.42 of the book.

Problem statement

Consider the following rational-input decision version of minimum-volume optimization. The input is $X \in \mathbb{Q}^{m\times n}$, an integer $2 \leq r \leq \min(m, n)$, and $\tau \in \mathbb{Q}_{\geq 0}$, with the promise that $\operatorname{rank}(X) = r$ and that there exist real factors $X = W_\star H_\star$ feasible for (MV) with $H_\star$ satisfying SSC. These factors are not supplied. Decide whether

$$\exists W \in \mathbb{R}^{m\times r},\, H \in \mathbb{R}_+^{r\times n} : \quad X = WH, \quad e_r^T H = e_n^T, \quad \det(W^TW) \leq \tau.$$

Does a polynomial-time algorithm exist, with time measured in the total binary input length? Randomization is allowed with success probability at least 2/3 on every promised input. The algorithm must run in polynomial time on all inputs but need only answer correctly on promised inputs. Checking the SSC promise is not part of the task. This decision formulation fixes the computational model of the book's polynomial-solvability conjecture; it does not assert an equivalence with exact factor recovery.

References

Gillis, *Nonnegative Matrix Factorization*, §4.3.3.6, pp. 148–149, with Definitions 4.15 and 4.42 and Theorem 4.43; author-hosted book. Barbarino, Gillis, and Saha, *Robustness of Minimum-Volume Nonnegative Matrix Factorization under an Expanded Sufficiently Scattered Condition*, §5, final research question.

Status check — 2026-09-10

Rechecked Barbarino–Gillis–Saha, §5, and searched for minimum-volume NMF complexity under sufficient scattering. The paper still asks about polynomial solvability even under its stronger p-SSC assumption; its robustness results assume a global optimum. No algorithm for the displayed rational decision promise was located. Gillis–Luce studies checking SSC itself, which is excluded from this promised-input task.